1. Mixed-component polynomial index calculus on a quartic twist

22 October 2026. Canonical quadratic twist at p=37, b0=2 over Fp^4, prime subgroup r=1,873,633, one public target Q=[271]G, seed zero. Frozen protocol: QUARTIC_TWIST_LINE_PROTOCOL.md. Native records and independent audits: MIXED_POLYNOMIAL_REPORT.md and MIXED_POLYNOMIAL_RUNS.md. This PDF is the rendered results summary; the Markdown report contains the complete evidence links, algebra and requirement table.

1.1. Mechanism

The transported Frobenius and cubic automorphism preserve the L0=Fp and L2=t^2Fp abscissa lines. Their combined signed action has order 24 and abscissa orbits of size 12. Thus 36 physical L0 plus 12 physical L2 factors give four log columns. Adding L2 raises physical relation yield; orbit folding reduces the number of independent log unknowns. Transforming one entire relation only gives a covariant row for the transformed residual, so symmetry does not create independent rows at fixed base and residual.

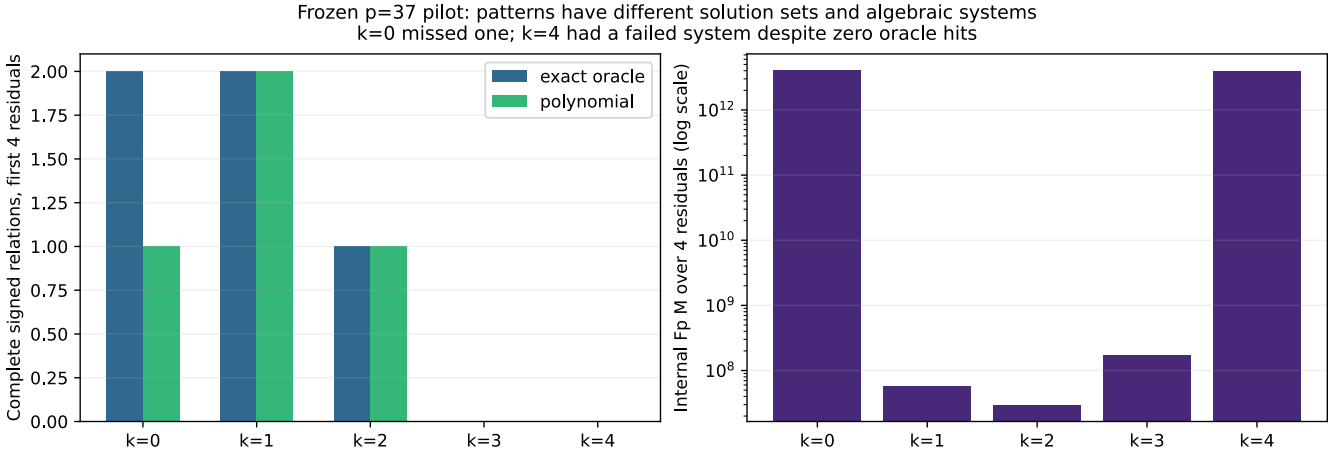
With one signed L2 point peeled, k=1 uses the quartic summation polynomial S4 in the three symmetric L0 abscissae. Four verified k=1 relations from eight frozen residuals have anchored rank four, solving three factor logs and the target log. Flat and cubic-only encodings would have 48 and 16 anchored unknowns for the same physical base, so four rows are insufficient.

1.2. Polynomial systems and pilot

For a=0, e1=x+y and e2=xy, the cubic equation at residual abscissa z is:

$$S3 = z^2(e1^2-4e2) - z(2e1e2+4b) + e2^2-4be1.$$

The k=2 arm explicitly scans all p^2 symmetric pairs after peeling two signed L2 points. The k=1 and k=3 arms use `S4 = Res_u(S3(x1,x2,u), S3(x3,z,u))`, interpolated and checked at fresh field points. The quadratic–quadratic Sylvester resultant identity is `(a2b0-a0b2)^2-(a2b1-a1b2)(a1b0-a0b1)`. Identity certificates IDC1h353e9e2256385dd8 and IDC1hb4b71bf7bd6cba4e passed 32 sealed and 32 independent checks each; these finite evaluations do not replace a formal model proof.

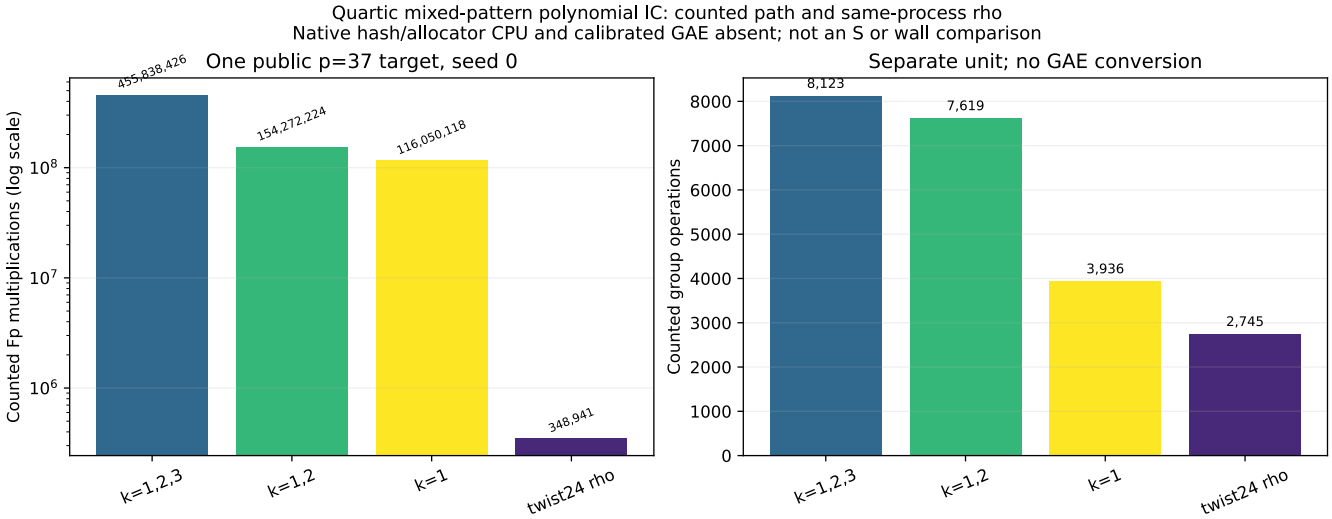


The first four frozen residuals contain two k=0, two k=1, one k=2, zero k=3 and zero k=4 signed relations. An independent scalar-keyed oracle enumerated all 1,873,633 subgroup logs. k=1,2,3 returned complete pattern-specific sets. k=0 missed one relation; k=4 had a failed system despite an empty reference set. The first generic k=2 Macaulay solver failed on all 1,056 systems; a complete Fp^2 scan repaired that arm. The degree-30 and degree-31 S5 retries on residual 2 still returned “border unreachable at this degree.” The native S5 arm remains incomplete.

1.3. One-target counted path

The producer reconstructs the curve, 48-factor base, twist eigenvalues, four-column map and S4 interpolation; processes the frozen residual stream; retains rank-raising rows; solves and verifies every factor log and the target. It never calls the exact pair oracle or reads the known target scalar during relation construction or matrix solving. Input fixture creation is reported separately. The same-process strongest twist24 rho recovers the same target.

Patterns	Queries	Rank	Counted IC Fp M	IC group ops
k=1,2,3	7	4	455,838,426	8,123
k=1,2	7	4	154,272,224	7,619
k=1	8	4	116,050,118	3,936
twist24 rho	672 steps	target	348,941	2,745



The k=1 path attempted 192 S4 systems, produced four distinct signed relations, reached anchored rank four and recovered 271. The independent auditor reconstructed all processed signed sets, orbit coefficients and matrix, then verified every physical factor log, target and rho replay. Final-source A/A records match in all non-timing fields. Pruning k=2,3 decreased counted IC field work 3.928-fold on this one target; this is an adaptive observation, not a result for other curves or targets.

1.4. Cost boundary and next checks

The k=1 IC uses 332.6 times rho's separately counted Fp multiplications and 1.434 times its counted group operations. A combined S or wall-speed ratio is not admitted: solver hash/tree and allocator CPU, peak bytes, rho partition/table CPU, calibrated GAE and isolated intervals are missing. No same-input msolve or Magma comparison was run. The result is therefore outside priced leaderboard tables A–C and benchmarks/README.md. The component panel links this report without promoting a speed claim.

Under a uniform-residual heuristic, k=1 yields O(1) hits per residual when both line sizes are O(p) and r is $O(p^4)$, while the fold has O(p) columns. Explicitly peeling O(p) signed L2 points per query gives $O(p^2)=O(r^{\wedge}(1/2))$ field work if S4 solve cost and row independence stay controlled. Those assumptions are unverified; this is no lower bound on other solvers.

Next: complete S5 across the tagged union, export identical systems to msolve, price native hash/allocator and rho CPU and isolated GAE bands, and run a frozen p=73/p=97 panel with independent replay. The related problem record is problems/quartic-twist-polynomial-ic-pricing.md.